Early Prediction of Complicated Pediatric Appendicitis Using Temporal Clinical Progression and Causal Risk Modelling

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***Abstract* - Pediatric appendicitis is one of the groups of the most frequent surgical crises in children, the misdiagnosis of which results in such complications perforation, the appearance of abscesses, and the prolonged postoperative period. The insufficient nature of specific symptoms and the dynamic progression of the disease are related to problematic early diagnosis of the complicated appendicitis. The existing diagnostic tools primarily rely on the single clinical scores, admission laboratory results, and imaging outcomes, which are unable to capture the early inflammatory development and can be analyzed causally. The given article is an innovative approach that will assist in forecasting early complications of pediatric appendicitis in the context of temporal clinical progression modeling and probabilistic causal risk analysis. The Regensburg Pediatric Appendicitis Data of demographic, clinical, laboratory and ultrasound-collected characteristics is structured into inflammatory dynamics according to time to reflect the changing trends of predicting complicated appendicitis. Experimental evidence has shown that the temporal and synthesized physiological attributes are significantly better predictors of complex appendicitis compared to the conventional fixed machine learning models. Also, causal risk modeling can provide interpretable clinical processes to severity of disease, enhancing clinical trust and decision-making.**

***Keywords*:** Pediatric Appendicitis, Complicated Appendicitis, Temporal Clinical Modeling, Causal Risk Analysis, Machine Learning, Clinical Decision Support Inflammatory Markers.

1. INTRODUCTION

Acute appendicitis (AA): AA is the most common surgical emergency in the pediatric population of the world and poses a significant clinical and economic challenge to the health care systems. The disease entails acute inflammatory condition of the vermiform appendix, which is usually caused by luminal obstruction and subsequent bacterial proliferation, tissue ischemia, and in case of untreated disease, gangrenous evolution or perforation. In pediatric practice, the risk of untimely diagnosis is incredibly high. In contrast to adult population, children, especially infants and toddlers, have a swift physiological transition of simple inflammation to complicated appendicitis (CA) that involves perforation, local abscesses, or generalized peritonitis. The clinical problem is further enhanced by the fact that the pediatric appendix has a thinner wall and also a less established omentum which greatly limits the capability of the body to contain infection and thus peritonitis develops rapidly as in children which is higher than in adults.Although the art of diagnostic imaging and laboratory procedures has been perfected over decades, the so-called gold standard of diagnosis has been just out of reach during the first stages of the disease. Nonspecific clinical manifestations are common in the case of pediatric patients, including generalized abdominal pain, irritability, nausea, low grade pyrexia, which are reminiscent of a variety of self-limiting gastrointestinal diseases. Moreover, the age-associated communication barrier will frequently hinder the precise reporting of the pain distribution or the symptom onset, compelling clinical workers to use a subjective physical examination. Laboratory inflammatory biomarkers, including the White Blood Cell (WBC) count and C-reactive protein (CRP) levels, are already combined with clinical scoring tools, such as the Alvarado score or the Pediatric Appendicitis Score (PAS) as standard diagnostic practices. Although they are a structured way of conducting triage, these tools are basically constrained by their immutability. They give a cross-sectional snapshot of the situation of a patient at the time of admission but do not take into consideration the speed of inflammatory change, which can be a more effective predictor of impending perforation than any single lab value.

1.1 Research Gap

Critical review of the existing clinical practices as well as the existing literature on the use of Machine Learning (ML) shows that some fundamental research gaps exist that would render the seamless incorporation of AI into the surgical theater impossible. The first of the limitations is the lack of temporal awareness. The majority of modern ML models of appendicitis assume that the data of patients are temporally independent observations. They are based on feature of admission-time without taking into account that appendicitis is a biological process that develops instead of a point. Having reached a CRP of 40mg/L in a child who has only been symptomatic in 4 hours suggests a very different inflammatory aggressiveness as compared to the same number in a child who was symptomatic in 48 hours. Existing models do not model this velocity of progression, and as a result, predictive accuracy is significantly reduced in the clinically important early stages of appendicitis in the time period of greatest effectiveness of intervention.

Moreover, there has been a high level of interpretability and causality gap in existing frameworks. Whereas the modern explainable AI (XAI) methods give a ranking of the importance of features as to how a specific model focused on a particular variable, these are purely associative rankings. They do not have the ability to build a causal relationship between clinical variables and disease outcomes. A high stakes surgical environment necessitates clinicians having the probability score in addition to a reasoning engine and is able to perform counterfactual analysis.
Available literature lacks a mechanism to respond to What-if situations, including what is predicted to happen to the risk of perforation in case a patient is kept in an active observation state

instead of operating. Finally, there is a repetitive dependency on the high-radiation imaging.

Numerous predictive models with high accuracy depend on the results of Computed Tomography (CT). The ionizing radiation of the CT scans is of major concern in the pediatric environment where malignancy is a risk that has long-term effects. There exists a serious deficiency in strong constructs that are able to attain high sensitivity of complex cases by utilizing non-invasive ultrasound and primary laboratory information only.

1.2 Motivation

The urgency of the proposed research is based on the fact that there is a conflict of priorities in pediatric emergency care based on the necessity to prevent life-threatening complications by means of timely intervention and to reduce the negative appendectomy rates (NAR) to avoid unnecessary surgical trauma and medical expenses. At the moment, NAR is still as high as 10-15% across most pediatric facilities, and the percentage of perforation is still regnally high as a result of diagnostic indecision. This is an indication that the existing paradigm of diagnosis has not been able to offer the accuracy that is required in order to have individualized surgery timing.

Causal Inference and Conformal Prediction are the latest concepts to be introduced into the wider body of AI, and this has provided a timely chance to solve these dilemmas as applied to appendicitis. With the incorporation of a causal logic we can advance beyond the predictions in a black box towards a more mechanism sensitive prediction system that is compatible with surgical reasoning. Besides, using conformal prediction, we can deliver safety-calibrated outputs. Massive consequences of being wrong in pediatrics. We are driven by the desirability of building a model that understands when it is uncertain. Instead of compelling a model to make an ambiguous binary prediction in the gray cases, we hope to enable the model to express its own lack of certainty, thus putting in the red flag on the particular patients that requires urgent senior clinical attention. This paradigm shift of the previously considered "static classification" to the modern-day temporal risk trajectory is necessary to modernize the pediatric emergency department and focus on the machine learning implementation on the concept of the Safety-First approach.

1.3 Key Contribution

This paper presents a new multi-layered model called Causal-Aware Conformal Prediction (CACP) to make early prediction of complex pediatric appendicitis. The Temporal Velocity Engineering is the initial significant input of this work. We suggest a way of organizing clinical, laboratory, and ultrasound-received data into data that can be derived as time dependent physiological indicators. A Velocity Score, which normalizes the inflammatory markers with the period of the symptoms, gives us a dynamic picture of the disease, something none of the earlier models with its fixed parameters gives.

The second contribution is the Probabilistic Causal Risk Discovery. The proposed framework goes beyond correlation-based feature importance and employs Structural Causal Modelling (SCM), to determine the interpretable pathways to complicated appendicitis. This can be used to estimate risks with mechanism awareness that gives the clinician a mathematical foundation on which to base their decision during surgery. Thirdly, we incorporate a Conformal Calibration Layer that ensures there is a given error rate. This will make sure the model gives a set of prediction (e.g. this patient might be Simple or Complicated) instead of a single prediction in the case where the data is mathematically ambiguous. The contribution plays a crucial role in clinical trust because it enables the AI to efficiently respond by saying I don't know which is the most significant safety aspect of medical AI. Finally, the design of the methodology is to be applied without any obligatory use of the CT images to facilitate radiation-free pediatric care and cost-effective.

1.4 Organization of the Paper

The rest of this research paper is structured such that it is able to offer a detailed and technically sound validation of the proposed framework. Section II: Literature Review is a thorough review of the 20 state-of-the-art articles in the field of pediatric appendicitis, which shows how the clinical scores evolved to be explained by AI and what the specific limitations of each method are. Section III: Proposed Methodology provides a description of the Regensburg Pediatric Appendicitis Dataset and explains the mathematical implementation of our five fundamental modules, which are temporal feature engineering, causal discovery, and conformal calibration. The individual algorithms that are employed in risk trajectory projection are also presented in this section. Section IV: Experimental Results and Discussion provide the performance of our framework as opposed to standard benchmarks using the following metrics: AUC-ROC, Brier Score, and Prediction Set coverage. Our causal findings have implications on clinical implications which we also discuss in detail. Lastly, Section V: Conclusion and Future Work overview the most important findings and provides the roadmap towards future prospective multi-center validation and real-time clinical integration.

2. PROBLEM STATEMENT

The current state of the diagnostic management of pediatric appendicitis is a kind of fundamental incompatibility between the dynamic pathology of the disease and the rather conceptually static diagnostic frames. Though acute appendicitis is a progressive inflammatory disease, the existing clinical decision-support tools use cross-sectional information; one-point lab results and admission-time imaging, that does not reflect the velocity of infection over time. It is this temporal blindness that results in a serious clinical failure the failure to discriminate between high-velocity aggressive inflammation and slow-progressing cases.

Moreover, the predictive models that are currently available are mostly associative, but not causal. On the one hand, they are capable of establishing correlations between biomarkers (e.g., levels of CRP) and the severity of the disease; however, causally, they are not as interpretable as they could be counterfactual reasoning. Therefore, the clinicians cannot mathematically estimate the direct impact of a particular wait-time or intervention on the likelihood of perforation. Lastly, the lack of uncertainty quantification in deterministic models compels binary classification of cases of uncertainty. In pediatrics where the price of a false negative is a life-threatening complication, lack of ability of a model to express mathematical doubt is a serious safety threat. The purpose of the study is to address these gaps by coming up with a framework to combine temporal progression modeling, structural causal discovery, and safety-calibrated uncertainty quantification.

3. RELATED WORKS

Recent articles have also used machine learning (ML) algorithms to support the diagnosis and treatment of pediatric appendicitis using a mixture of heterogeneous clinical data. As demonstrated by Navaei et al. [12], a composite of clinical, lab, and imaging features in the supervised ML models proved to be far superior to the past heuristic scoring systems based on diagnostic accuracy. Similarly, Pati et al. [13] employed the methods of ensemble learning, including the majority voting and weighted averaging, to predict various results associated with appendicitis which showed a greater strength and a better quality of classification. Even though this kind of data-based methods enhances the credibility of diagnostic data, they are more inclined toward the detection rather than early stratification of the disease severity.

Besides using the ML-based diagnosis, other studies have also examined the use of biological markers to distinguish the uncomplicated and complicated appendicitis. According to Ha et al. [14], blood-based inflammatory biomarkers identified that C-reactive protein and leukocyte-derived indices were significant predictors of complicated appendicitis in children. Based on the same biological style, Liu et al. [15] discovered two possible microbial states with clinical indications and severity, suggesting the potential to apply profiles of gut microbiome in disease progression.

The use of artificial intelligence techniques that concentrate on imaging and can be interpreted have become common as well. The machine learning models proposed by Marcinkevics et al. [16] prioritize interpretability and clinician intervention, which overcomes the black-box of traditional AI models, on the basis of ultrasonography. In conjunction with this, Abu-Ashour et al. [17] have demonstrated that natural language processing-based free-text operative and ultrasound report labeling can automatically grade appendicitis, to enable scalable clinical decision support to be realized. Nevertheless, all of these methods remain dependent on the quality of imaging and experience of the operator.

The problems of early severity prediction are also underlined in treatment-oriented studies. Conservative management that includes the use of antibiotics has proven to yield positive outcomes in the selected groups of pediatrics as Tan et al. [18] and Monti et al. [19] revealed, and uncomplicated cases need to be defined appropriately. On the other hand, as Goto et al. [20] have shown, false-negative score of appendicitis is possible because of the presence of antibiotic treatment, which occurs prior to diagnostics that can delay the further intervention. All these results highlight the evident gap in the necessity to possess a temporally aware, causally superlative forecast structure that can be applied to make priori and individualized treatment judgments.

Recent research in pediatric appendicitis has increasingly focused on improving diagnostic accuracy and severity prediction using machine learning (ML), biological markers, and imaging techniques. Traditional scoring systems such as the Pediatric Appendicitis Score (PAS) are often limited by subjectivity and lower predictive performance, which has led to the adoption of data-driven approaches.

Navaei et al. [12] proposed a hybrid diagnostic framework that integrates clinical, laboratory, and sonographic features to improve the accuracy of appendicitis detection in children. Their study demonstrated that combining heterogeneous data sources significantly enhances predictive performance compared to conventional scoring systems, especially in resource-constrained environments. Similarly, Pati et al. [13] conducted a comparative analysis of multiple ML models, including Random Forest, XGBoost, and Logistic Regression, using clinical and laboratory data. Their results showed that ensemble learning techniques, such as majority voting and weighted averaging, provide improved classification accuracy and robustness in predicting appendicitis outcomes.

In another study, a large cohort-based ML pipeline [3] utilized imputation and upsampling techniques to predict multiple severity levels of appendicitis. This approach improved preoperative classification but was highly dependent on preprocessing strategies. Additionally, research involving 551 pediatric patients [2] applied ML algorithms to reduce negative appendectomies; however, the model suffered from low specificity and required further external validation. A retrospective study [1] comparing clinical features and imaging findings in young children identified key predictors for perforated appendicitis, though the small sample size limited generalizability.

Beyond ML models, several studies have explored the role of inflammatory biomarkers in severity differentiation. Ha et al. [14] demonstrated that C-reactive protein (CRP) and leukocyte indices are strong indicators of complicated appendicitis. Supporting this, another study [11] analyzed multiple inflammatory markers, including WBC, PCT, TNF-α, and IL-6, and established their correlation with disease severity and recovery outcomes.

Similarly, research focused on diagnostic utility [8] confirmed that biomarkers such as CRP and neutrophil counts can effectively distinguish between complicated and uncomplicated cases, although such approaches are limited by their reliance solely on laboratory data.Recent advancements have also explored microbiome-based analysis. Liu et al. [15] identified distinct microbial patterns associated with appendicitis severity, providing new insights into disease progression. A related study [9] classified intra-abdominal samples based on microbial composition and linked these patterns with clinical severity. However, both studies were constrained by small sample sizes and limited clinical integration.

The application of explainable AI and imaging-based models has further improved clinical interpretability. Marcinkevics et al. [16] introduced concept bottleneck models using ultrasound images, enabling interpretable predictions that allow clinicians to understand the reasoning behind model outputs. Similarly, Abu-Ashour et al. [17] utilized natural language processing techniques to extract structured information from free-text clinical and ultrasound reports, facilitating automated grading of appendicitis. Despite these advantages, such models are highly dependent on imaging quality and operator expertise.

Treatment-oriented studies emphasize the importance of early and accurate severity prediction. Tan et al. [18] and Monti et al. [19] evaluated conservative management strategies using antibiotics and found them effective in selected uncomplicated cases. However, Goto et al. [20] highlighted that prior antibiotic administration can lead to false-negative diagnostic results, delaying necessary surgical intervention. Additionally, studies [4] have shown that prediagnostic antibiotic use can affect the reliability of scoring systems such as PAS.

Overall, existing literature demonstrates significant progress in improving appendicitis diagnosis using ML, biomarkers, and imaging techniques. However, most approaches focus primarily on diagnosis rather than early severity prediction. Furthermore, many studies rely on single-modal data and lack temporal analysis, limiting their ability to capture disease progression. These limitations highlight the need for a temporally aware, multi-modal, and interpretable predictive framework that can support early risk stratification and personalized treatment decisions in pediatric appendicitis.

Ref	Dataset	Methodology	Application	Limitations
[1]	2 centers with 72 children less than 3 (March 2018- May 2024).	The comparison of the clinical features and imaging (Ultrasound/CT) retrospectively to make a comparison between the perforated and non-perforated groups.	Identification of infants- and toddler-specific predictors of clinical characteristics and perforation to improve early diagnosis.	Small size of sample (72 patients) which might not allow generalizing the findings to larger population of pediatric population.
[2]	Sample of 551 suspected pediatric age patients with acute appendicitis who underwent surgical operations.	Compared three ML algorithms (Random Forest, XGBoost, and Logistic Regression) based on clinical and laboratory characteristics.	Training an explainable ML model to reduce negative appendectomies (estimated reduce by 17%).	The inability to achieve high model specificity (0.17) was one of the constraints, and it is acknowledged in the research that the model should be externally validated.
[3]	Cohort (1980 children) of 1, 980 children with emergency appendectomy (2014-2021).	ML pipeline based on imputation and upsampling to forecast five different severity grades of appendicitis.	Improving the preoperative diagnosis and appropriate classification of severity (non-perforated, generalized peritonitis).	The performance also depends on the certain mixture of imputation and class-balancing methods.
[4]	Single-centre retrospective cohort pediatric.	Correlation study of prediagnostic administration of antibiotics and false-negative Pediatric Appendicitis Scores (PAS).	Identifying the factors of false-negative PAS outcomes that can delay the diagnosis and may result in complications.	Single-center design could limit generalization of findings and results to other health care settings or protocols.
[5]	2,100 children; 103 reports vs. AI in EMR Free-text data.	ChatGPT-4 comparison with data extraction and structuring with human collectors.	Large scale data science and grading Automation of naming of unstructured operative and ultrasound reports.	AI performance is conditional upon timely engineering and professional management of complex medical peculiarities is still required.
[6]	Ultrasound (US) images, clinical and laboratory outcomes of 579 Pediatric patients of 1709.	Multiview US pictures and missing concept set was made into Concept Bottleneck Models (CBM).	Creating image-based models that are interpretable and manipulateable on prediction of diagnosis, management and severity of images using imaging.	The model may be able to predict variables variably by using the incomplete sets of concepts in training.
[7]	The number of suspected appendicitis children and adolescents (0-18 years) was 430.	Weighted averaging/majority voting of multiple classifiers ensemble learning.	Predicting target variables to improve clinical urgency assessment and medical judgment tailoring.	Lack of an agreement between the international community on the management suggestions limits clinical normalization of the models.

Ref	Dataset	Methodology	Application	Limitations
[8]	Children (Pediatric patients with acute abdominal pain: Acute Appendicitis).	Diagnostic utility Diagnostic utility analysis of blood biomarkers Diagnostic utility to discriminate between types of disease Diagnostic utility (CRP, neutrophil counts).	Surgical intervention-Differentiation between complicated (CA) and uncomplicated appendicitis (UA).	Almost entirely interested in blood fields; does not have an interface to high-end ML or multi-modal image data
[9]	74 intra-abdominal pus samples in children between the ages 3 and 15.	The classification of pathology as per microbial condition (dysbiosis) of samples.	Associating definite microbial compositions and clinical signs plus disease severity (Simple vs. Complicated).	The sample size (74) was weak, and this limitation restricted itself to the intra-abdominal pus that may not be the full range of the clinical picture.
[10]	Pediatric clinical information.	ML model that uses five explainable AI (XAI) that were used to evaluate variables including Alvarado score and CRP.	Whether complicated or uncomplicated appendicitis can be predicted preoperatively to aid in clinical risk stratification	Retrospective design; may have institutional bias; may require feature quality; may be externally validated.
[11]	150 patients that had confirmed AA and 150 healthy controls of the pediatric age.	The use of retrospective cohort study based on the inflammatory markers (WBC, PCT, CRP, TNF-a, IL-6) preoperative and the Alvarado scores.	Establishes the correlation between the preoperative markers and the severity of the disease, recovery and pain.	Lack of generalizability; small sample size The study is retrospective.
[12]	Abdominal pain in young patients 2-14 years.	Developing a hybrid clinical-biological score, which is a combination of clinical, biological and sonographic data.	It was filled in with the identification of key predictive factors of AA so as to enhance the accuracy of diagnosis in a resource constrained environment.	Specific to a single population that will require external validation.
[13]	The AA patients used in the study (n=1,132, 2012-2022) were divided into training, testing, and validation sets.	Comparison of 6 machine learning (ML) algorithms with respect to clinical and laboratory features of significant clinical and laboratory features.	Establishes the classification of severity of acute appendicitis (perforated and non-perforated) according to the simple clinical parameters.	Historical data can be biased in terms of selection; the performance of the model can be influenced by the quality of the historical records.
[14]	Identifies the acute appendicitis according to its severity (perforated and non-perforated) according to simple clinical indicators.	Prospective analysis Prospective analysis (Pediatric Appendicitis Score PAS) and (CRP levels).	Recent discoveries of valid laboratory biomarkers that can help establish earlier differentiation between mild and severe appendicitis in children and their use in surgical decision making.	Single-centered research; small sample size; the results might need extrinsic confirmation.

Ref	Dataset	Methodology	Application	Limitations
[15]	Pediatric patients with acute abdominal pain (Acute Appendicitis).	Microbiome analysis to identify distinct microbial states in pediatric acute appendicitis. Statistical association analysis between microbial patterns and clinical severity markers was performed.	Identification of microbiological biomarkers associated with appendicitis severity in children to improve disease stratification and understanding of pathophysiology.	Single-cohort observational design; limited generalizability across diverse populations; lack of external validation; retrospective clinical data integration.
[16]	74 intra-abdominal pediatric pus samples in the age 3-15.	Pathological classification according to microbial condition (dysbiosis) of samples.	Association between microbial composition and clinical phenotypes and disease severity (Simple vs. Complicated).	Weak sample size (74) restricted itself to intra-abdominal pus which might not be the entire spectrum of the clinical picture.
[17]	Clinical data from pediatric patients.	ML model based on five explainable AI (XAI) methods to assess variables such as Alvarado score and CRP.	One hundred and forty percent accuracy (94 percent of severity, 89 percent of management) to minimize diagnostic delays.	Depending on clinical indicators that may go unnoticed, or delayed by the patients, until the hospital is visited.
[18]	150 patients with established AA (pediatric) and 150 healthy controls..	Retrospective cohort study involving the preoperative inflammatory markers (WBC, PCT, CRP, TNF-a, IL-6) and Alvarado scores.	Determines the relationship between the preoperative markers and the severity of the disease, recovery, and pain.	Limitations to generalizability; small sample size; The study is retrospective.
[19]	Pediatric patients with acute appendicitis managed conservatively in a single tertiary care center (monocentric study).	Retrospective evaluation of outcomes following non-operative (conservative) management of pediatric appendicitis. Analysis and complication and clinical follow-up were performed.	Assessment of feasibility, safety, and effectiveness of conservative (antibiotic-based) treatment strategies in children with acute appendicitis.	Limitations to generalizability; small sample size; The study is retrospective.
[20]	One thousand one hundred and thirty-two (1,132) AA patients (2012-2022) in training, testing and validation sets.	Comparison of 6 machine learning (ML) algorithms with substantial clinical and laboratory features.	Evaluation of how prior antibiotic administration affects diagnostic accuracy and contributes to false-negative appendicitis scores in pediatric patients.	The retrospective data can be subjective to selection bias; the quality of the historical records will determine the quality of model performance.

4. METHODOLOGY

The proposed framework addresses two critical weaknesses of existing pediatric appendicitis prediction frameworks, which include the absence of time modeling of the disease process and the absence of direct quantification of uncertainty. Most of the past approaches rely on clinical snapshots and provide deterministic predictions which may be unsafe under uncertain clinical conditions. In an attempt to address this, Causal-Aware Conformal Prediction (CACP) framework is introduced, which includes temporal dynamics of symptoms, structural causal modeling and distribution free calibration of uncertainty. The model is a combination of causal inference and conformal prediction to. provide sets of risks that are certified to be safe, as compared to hard binary decisions which permits clinicians to postpone an intervention in situations whereby, they feel insufficiently certain. The whole working process begins with Temporal Velocity Feature Engineering where inflammatory characteristics have been normalized based on the duration of the symptoms in order to capture the rate of disease progression. Structural Causal Discovery Module is then applied to learn clinically useful causal pathways and select spurious relationships. Gradient-boosting models are used to analyze the filtered properties with a weighted combination of gradient-boosting models. The resultant system gives a curve of the risk trajectory, which can be visually represented to show how risk of perforation varies with time, and the counterfactual analysis that can approximate the impact of wait time on clinical outcomes

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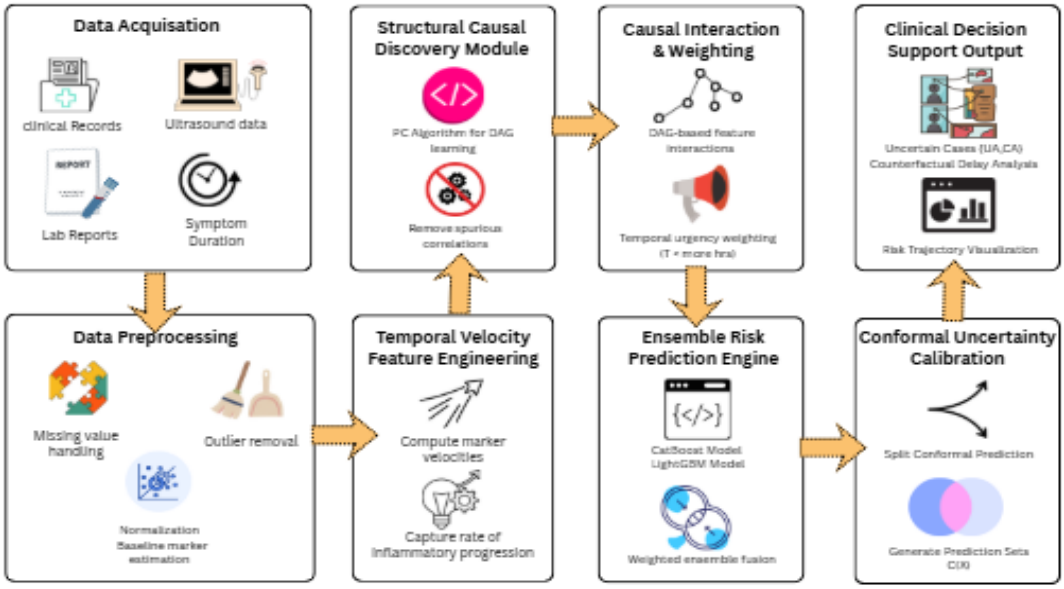

Fig 1 System Architecture

4.1. System Overview

The suggested system is a clinical decision support system using artificial intelligence, the proposal is that this system will help detect complicated appendicitis (CA) in children at an early age. The system combines the temporal modeling, structural and ensemble machine learning and uncertainty-aware prediction into a single pipeline.The key objective is to detect the high-risk cases before they become clinically deteriorated. Unlike other conventional correlation based classifiers, the system employs time relevant characteristics of velocity (physiologically) and causal feature filtering to improve robustness and interpretability.

4.2. Clinical Data Architecture

The system uses the Regensburg Pediatric Appendicitis Dataset (RPAD), a publicly available dataset and a clinically proven dataset of approximately 750 children who were tested because of suspected acute appendicitis. In every patient record, there is:

Demographic characteristics: age, sex.

Clinical signs: the shifting of the pain, the vomiting, fever.Laboratory indicators: the white blood cell count (WBC), the percentage of neutrophils, C-reactive protein (CRP), the number of platelets.Imaging: indicators obtained through ultrasound (appendix diameter, free fluid, wall thickening).

4.2.1.Preprocessing and Missing value treating

The majority of clinical data sets can be missing laboratory or imaging data.. Missing data are filled in with median imputation in eq.1.

$$x_{ij} = \begin{cases} \text{median}(x_j), & \text{if } x_{ij} \text{ is missing} \\ x_{ij}, & \text{otherwise} \end{cases} \quad (1)$$

There are dissimilar numeric ranges of the variables in the laboratory.

4.2.2 Feature Scaling

It has stable model learning and normalization using Z-score in eq.2.2

$$x_{ij}^{norm} = \frac{x_{ij} - \mu_j}{\sigma_j} \quad (2)$$

where mjand s are the mean and the standard deviation of feature j.

4.3 Temporal Clinical Progression Modeling

Appendicitis is in the active stage of development. The characteristics of time velocity are constructed to encode the evolution of inflammatory.

4.3.1 Construction of Features based on velocity

A temporal velocity feature of each inflammatory marker M_i is then calculated in eq.3.

$$V_{M_i} = \frac{M_i - M_i^{\text{baseline}}}{\ln(T+1)} \quad (3)$$

In which M_i is the measured marker value,M_i baseline" indicates a reference normal value of the symptom,Trep is the duration of the symptom (in hours).

4.3.2 Composite Temporal Indicators

Clinically related markers were grouped into physiological meaningful indices in eq.4.

$$\text{IPI} = \alpha \cdot V_{\text{CRP}} + \beta \cdot V_{\text{WBC}} + \gamma \cdot V_{\text{Neutrophils}} \quad (4)$$

a,b,g are coefficients obtained in the training phase of the model and IPI is the abbreviation of Inflammatory Progression Index.

4.4 Structural Causal Risk Modeling

A Structural Causal Model (SCM) is created to overcome the correlation-based modeling through the application of PC algorithm in eq.5.

Let:

$$G = (V, E) \quad (5)$$

where V,E are clinical variables and directed causal edges respectively.

4.4.1 Causal Feature Selection

Only variables whose causal relationships with outcome Y are proven to be valid in eq.6.

$$\text{Retained Features} = \{ X_i \mid X_i \rightarrow Y \in G \} \quad (6)$$

This guarantees clinical interpretability and minimizes predictors which are spurious.

4.5 Ensemble Risk Prediction Core

Core Prediction Ensemble This is an ensemble risk prediction technique that applies a large ensemble of component risk predictors to generate predictions.The final prediction is in eq.7.

$$\hat{y} = \sum_{k=1}^{K} w_k f_k(X) \quad (7)$$

wheref_k is the k th base learner (CatBoost or LightGBM), w_k is the resulting learnt weight and X is the causally filtered temporal feature *vector.*

4.6 Uncertainty Quantification via Conformal Prediction

Conformal prediction is used to enhance clinical safety.

4.6.1 Nonconformity Scores

The nonconformity scores are obtained with a set of calibration data called D not under investigation D_{cal}:The system is capable of learning to predict the set of observations it observed previously calculated in eq.8.

$$s_i = 1 - \hat{P}(Y_i \mid X_i) \quad (8)$$

4.6.2 Prediction Set Construction

The system can learn how to predict the set of observations which it has seen earlier.The Prediction set C(X)satisfies in eq.9.

$$P\big(Y \in C(X)\big) \geq 1 - \alpha \quad (9)$$

where ais is a risk tolerance that is defined by the user(e.g., 0.05).

The model gives both labels (UA, CA) in ambiguous cases, hence resulting in a real-time clinical re-evaluation a safety-oriented action that upholds the concept of pediatric decision-support.

Algorithm 1 Causal–Velocity Feature Engineering

Input: Clinical marker set
M = {CRP, WBC, Neutrophils, Lymphocytes}
Symptom duration T(hours)
Base line reference values $M_{baseline}$.
Output: Increased time-causal feature X'
Initialize:
Feature vector $X' = \emptyset$
Temporal weight Wt
1: Given every clinical marker $M_i \in M$ do
2: Computation of temporal velocity feature.
3: End For
4: Develop composite inflammatory index.
IPI=αVCRP+βVWBC+γVNeutrophils
5: Develop causal interaction feature with the help of DAG.
6: Use temporal urgency weighting.
if T< 24 hours
W_t>1
else
W_t=1
7: Model improved feature vector.
X′=[Wt·VMi,Wt·Icausal, IPI]
8: Return X'

Algorithm 2 Safety-Calibrated Conformal Prediction (SCCP)

Input: Calibration dataset
$D_{cal} = \{(x_i, y_i)\}_{i=1}^{n}$
Test instance x_{test}
Confidence level α
Output: The prediction label→ {UA}, {CA}, or {UA, CA}
Initialize:
Nonconformity score set $S = \emptyset$
Quantile threshold $\hat{q}$
1: Let (x_i,y_i) represents any calibration sample.
2: Calculate predicted probability f $(x_i)_{(yi)}$
3: Calculate nonconformity score.
4: Add S_i to set S
5: End For
6: Compute quantile threshold
q = Quantile(1−α)(1+1/n)(S)

7: Considering all possible labels $y \in \{UA, CA\}$do
8: Compute $1 - \hat{f}(x_{test})_y$
if 1−f (x test) ≤ q
Add y to prediction set $C(x_{test})$
9: End For
10: **if** |C(xtest)|=1
11: This high-confidence label is a single high-confidence label.
12: else
13: Return {UA, CA}
14: Instigate clinical reassessment.

5. RESULTS AND DISCUSSION

5.1 Experimental Setup

The proposed framework was evaluated using the Regensburg Pediatric Appendicitis Dataset (RPAD), comprising approximately 750 pediatric cases with demographic, clinical, laboratory, and imaging features. Data preprocessing included removal of duplicates, handling of missing values using median imputation for numerical variables and categorical encoding for discrete features, followed by standardization to ensure uniform scaling. Temporal feature engineering was performed by computing velocity-based indicators for key inflammatory markers such as C-reactive protein (CRP), white blood cell count (WBC), and neutrophil percentage, which were further aggregated into a composite Inflammatory Progression Index.

Causal feature selection was incorporated using a domain-informed Directed Acyclic Graph (DAG) to retain clinically relevant predictors. The dataset was split into training and testing sets using an 80:20 stratified sampling strategy to preserve class distribution. Multiple machine learning models, including CatBoost, LightGBM, XGBoost, Random Forest and Logistic Regression, were trained and evaluated. Model performance was assessed using accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC). Additionally, calibration analysis and conformal prediction were employed to evaluate probability reliability and quantify prediction uncertainty, ensuring robust and clinically interpretable outcomes.

5.2 Shortcomings of Raw Clinical Features

The statistical analysis of the raw clinical features is presented in **Fig. 2,** in which the visualization of the biomarkers, including C-reactive protein (CRP), white blood cell count (WBC), and neutrophil percentage, is performed in various classes of severity. One can clearly see that the distributions of these features are heavily overlapping between uncomplicated and complicated cases.

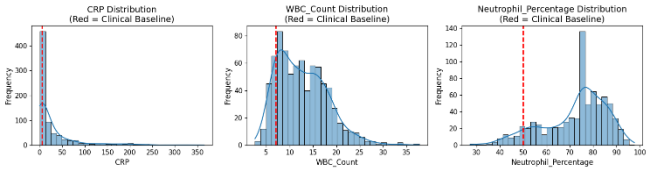

Fig 2. Marker vs Baseline Distribution

These overlaps imply that raw clinical markers do not allow enough discriminative power and especially in the early disease progression when the differences between symptoms are minor. It is a problem of feature space that there is no visible boundary between the decisions, so the classical classification models depend on the separability to make correct predictions.These results are a strong indication of why the methodological choice to have a temporal feature engineering since the disease is dynamic and not static. These results support greatly the methodological choice of including temporal feature engineering because the stagnant representations do not represent the changing character of the illness.

5.3 Study Effectiveness of Temporal Feature Engineering

In order to defeat the shortcomings of the fixed features, temporal changes were used to normalize features and make them more meaningful. The effect of this process can be seen in **Fig. 3** that shows that distributions of features that are normalized by time have lower variance and better cross-patient alignment in patients with varying symptom duration.

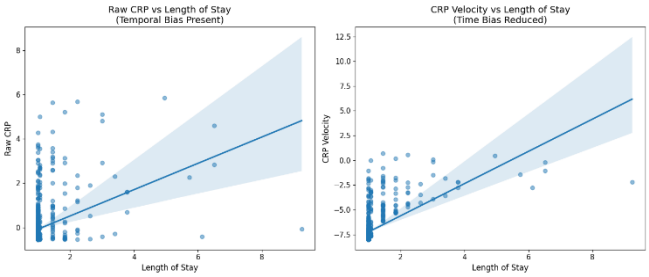

Fig 3. Before vs After Normalization Temporal Bias

Normalization across time is used to address discrepancies that would occur due to late visits to hospital and different times of observation. The model can understand clinical measurements in a consistent time context because it standardizes the time dimension.

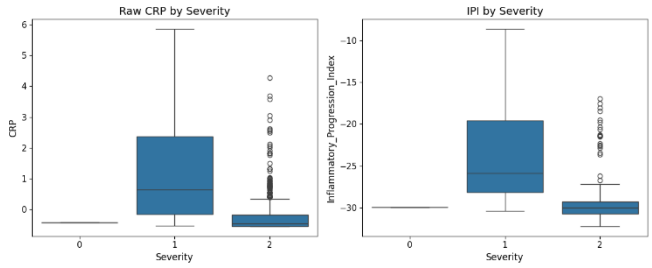

Fig.4 Raw CRP & IPI by Severity

The next improvement is seen in **Fig. 4** where the composite Inflammatory Progression Index (IPI) is presented. The IPI Unlike a single biomarker, the IPI combines several temporal signals into one which captures the rate and intensity of the progression of inflammation. The figure shows a better division of severity classes, which proves that feature fusion and temporal aggregation remarkably improve the discriminability of the classes.These findings confirm the hypothesis that disease complications can only be predicted correctly when the dynamics of time are incorporated.

5.4 Feature Relevancy and Correlation Analysis

Fig. 5 shows the correlation matrix which gives quantitative information on the relationship between the input feature and the target variable. It is noted that, temporally engineered features are more correlated with the outcome than raw clinical variables.

Fig 5. Correlation Matrix: Raw Markers and Temporal Velocities.

This indicates that not only does temporal processing enhance the representation of features, but also it enhances the feature relevance with reference to the task of prediction. Further, the interacting features show relevant interdependencies in the matrix, which implies that complex interactions are possible that could not be represented using a linear model.

The enhanced correlation framework favours the choice of improved ensemble models, which can exploit nonlinear relationships, as applied to the methodology. The 4 figure indicates more distinctiveness between the classes of severity which proves that the fusion of features and time is an important element that improves the class discriminability. The results confirm the hypothesis that temporal dynamics are necessary in making proper predictions of disease complications.

5.5 Validation of Causal Relationships

In order to achieve interpretability and clinical validity, we modelled causal relationships between variables with a DirectedAcyclic Graph (DAG), depicted in Fig. 6. Direct causal relationships between important clinical characteristics and disease outcomes are identified by the DAG and any spurious relationships are removed.

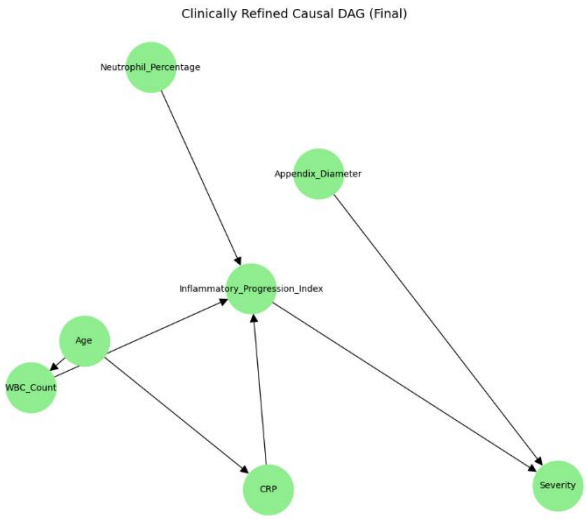

Fig 6. Clinically Refined Causal DAG

This filtering operation takes care of causal features and is used to filter out non-medically meaningful features, thus minimizing noise and enhancing generalization of the models. In addition, the strength of these causal relationships is measured by an edge-weighted graph in as shown in Fig.7.

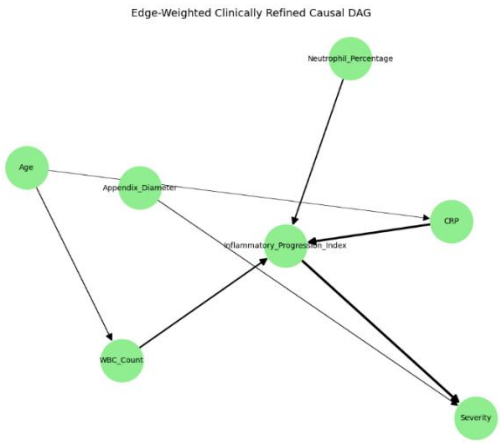

Fig 7. Causal Influence Graph: Edge-Weighted

The findings show that the inflammatory indicators over time are the most significant which concur with the known clinical information. This agreement between insights gained in the model and domain knowledge increases the confidence of the presented method.

5.6 Comparative Analysis of Machine Learning Model

The predictive performance of multiple machine learning models was evaluated using standard classification metrics. The results indicate that ensemble boosting methods outperform traditional approaches in capturing complex data patterns in Table.1.

Among all models, CatBoost achieved the best overall performance, with an accuracy of 0.9427 and the highest AUC of 0.9762, indicating strong capability in distinguishing between classes across different threshold values. It also maintained a good balance between precision (0.8000) and recall (0.8333), resulting in a solid F1-score of 0.8163. LightGBM showed comparable results with an AUC of 0.9756 and accuracy of 93.63%. XGBoost and Random Forest also performed well, with AUC values of 0.9696 and 0.9715, respectively, where Random Forest exhibited higher recall (0.9167) in identifying positive instances. In contrast, Logistic Regression yielded the lowest performance with an AUC of 0.8913 and accuracy of 89.81%, highlighting its limitation in modeling non-linear relationships. Overall, the findings confirm that boosting-based ensemble models, particularly CatBoost and LightGBM, are more effective for this dataset due to their ability to capture complex feature interactions.

Table.1. Model Comparison with Performance Metrics

S.No	Model	Accuracy	Precision	Recall	F1 Score	AUC
1	LightGBM	0.9363	0.7692	0.8333	0.8000	0.9756
2	Catboost	0.9427	0.8000	0.8333	0.8163	0.9762
3	XGBoost	0.9172	0.6774	0.8750	0.7636	0.9696
4	RandomForest	0.9299	0.7097	0.9167	0.8000	0.9715
5	LogisticRegression	0.8981	0.6538	0.7083	0.6800	0.8913

These findings corroborate the hypothesis that the data has complex, non-linear relationships within the data, and that the boosting-based ensemble models are the most appropriate in identifying these relationships.

5.7 Ensemble Model Robustness

In order to increase the stability of the prediction, the ensemble model was developed by combining the results of the most effective classifiers. The ensemble scored an AUC of 0.9827 which shows very high performance with respect to various data variations.

In spite of the fact that the ensemble fails to outperform the best individual model in terms of the highest AUC, it leads to better generalization and minimizes variance. The advantage of this trade-off in the real world is that consistency and reliability are paramount concerns. The findings confirm the methodological decision of using ensemble learning to meet a trade-off performance between different scenarios.

5.8 Risk Modeling Dynamically and Clinical Stratification

The dynamic response of the risk that has been predicted is plotted in Fig. 8, a 3D surface showing the risk change with respect to time and markers of inflammation. The surface has a nonlinear structure, which means that the risk accelerates at some definite levels of inflammation and time lag.

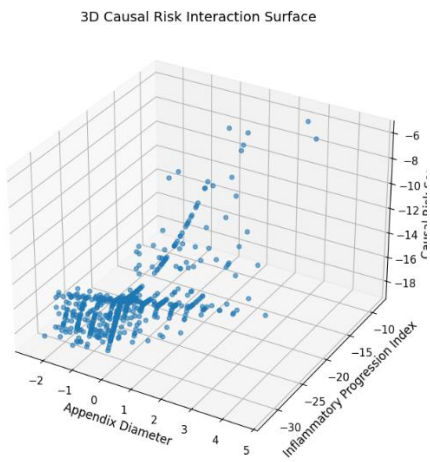

Fig.8. 3D Causal Risk Interaction Surface.

This nonlinearity demonstrates the necessity of prompt diagnosis and treatment, with minimal delays resulting in huge consequences of complication risk.

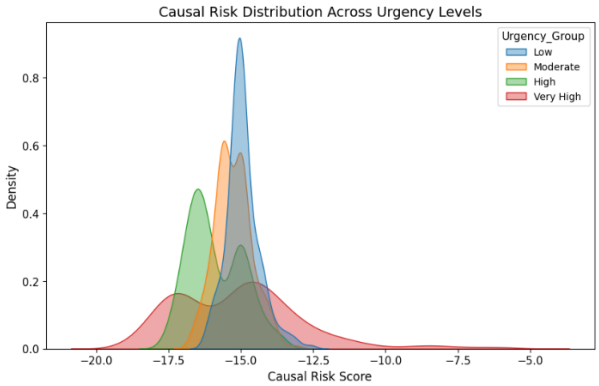

Fig 9. Risk Density by Urgency Quartiles

In addition, Fig. 9 shows the distribution of the anticipated risk among patients. The high-risk cases are concentrated at upper quartile, and the low-risk cases are at lower range, which is evidence of effective stratification.

5.9 Model Reliability and Uncertainty Quantification

One of the contributions of the proposed framework is the framework that incorporates the use of conformal prediction in estimating uncertainties. The model was used to predict 146 confident cases and 11 uncertain cases using a threshold of 0.35. This is done to make sure that predictions that are of low confidence are flagged thereby minimizing the chances of wrong clinical decisions. Compared to the traditional models, which deliver deterministic results, the suggested framework takes into consideration a safety-sensitive layer and is more applicable in critical medical systems.

5.10 Ablation Research and Elementary Competition

In order to evaluate the role of the temporal features, an ablation experiment was done by eliminating the temporal features of the model. The resultant AUC fell considerable at 0.9711 to 0.9558.

Such a significant drop proves that the time aspects represent an important aspect of the structure. It validates the assumption that the effective modeling of the disease progression is impossible with the help of the static features only, which proves the value of the suggested feature engineering strategy.

6. OBSERVATIONS

This section gives the essential findings based on the experimental findings, model outputs, and interpretability studies. These observations would help in gaining a better insight into the model behavior, feature significance, and clinical implications.

6.1 Importance of Features and Interpretability of the Model

Fig. 10 is a SHAP-based feature importance analysis that allows obtaining a global view of how the model makes its decision. All of these findings suggest that temporal markers of inflammation, including CRP progression rate and neutrophil percentage, have the largest impact on the prediction results.

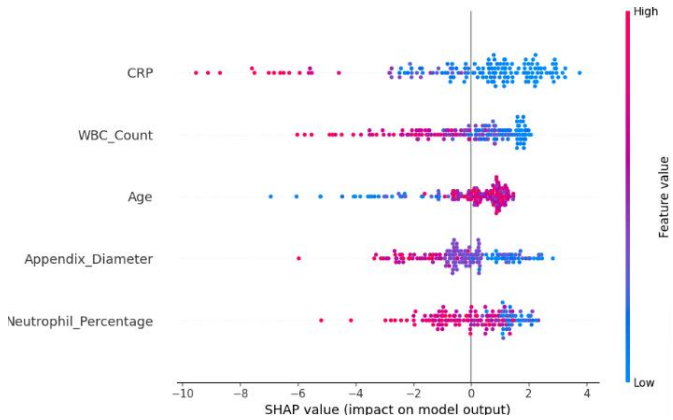

Fig.10. SHAP value Impact on model output

These engineered temporal variables (as opposed to raw features) record the magnitude and progression trends of features, and are therefore more informative. The difference between SHAP importance ranking and the causal relationships presented above (Fig. 6 and Fig. 7) supports the validity of the model.This correspondence shows that the model is not only correct but also clinically comprehensible, a necessary condition to work in practice.

6.2 Delay Sensitivity Analysis and Risk Trajectory

The forecasted risk profiles and delayed risk approximations are outlined in Fig. 11, where the characteristic instances point at the distinction between instant and delayed forecasts.

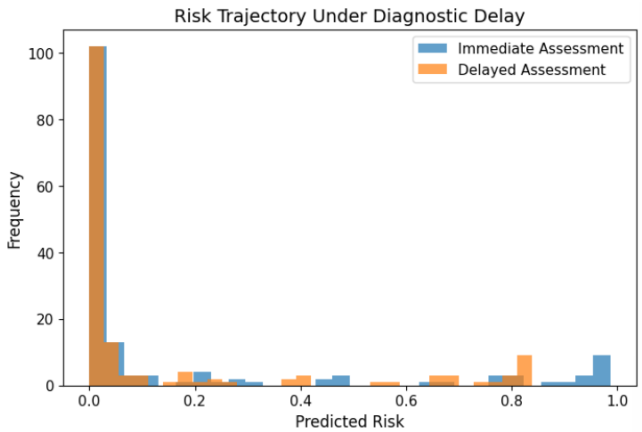

Fig.11. Risk Trajectory under Diagnostic Delay

The risk trajectory plot compares predicted risk under immediate versus delayed clinical assessment.Most cases cluster at low-risk values, indicating stable patients under both conditions.However, delayed assessment shows a noticeable shift toward higher risk values, suggesting potential disease progression.This highlights the importance of early diagnosis, as delays can significantly increase the likelihood of complications.

6.3 Uncertainty Analysis and Confidence Calibration.

Conformal prediction further provides an uncertainty-aware perspective by identifying cases where the model lacks sufficient confidence. A small proportion of instances (11 cases) are flagged as uncertain, while the majority (146 cases) are classified with high confidence. This behavior indicates that the model maintains strong predictive certainty for most patients while appropriately deferring ambiguous cases for further clinical evaluation, thereby enhancing

decision safety.

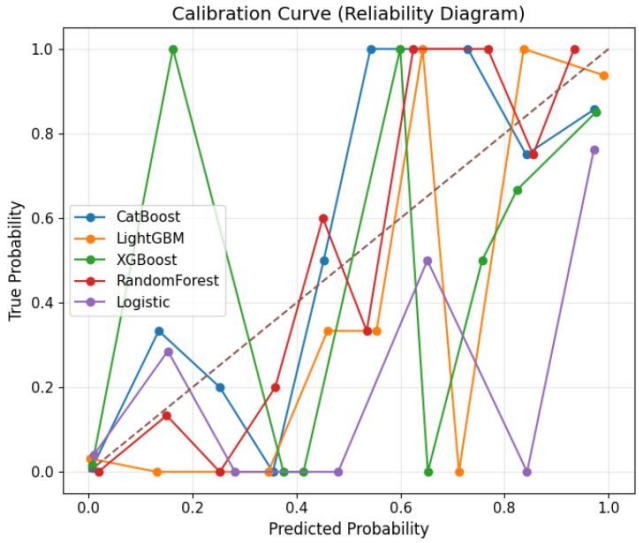

Fig.12. Calibration Curve

This implies that the model has a high degree of confidence in the majority of predictions, but it still retains the capability of indicating the cases of ambiguity. These predictions with uncertainty are especially useful in clinical practice, where a wrong choice may be very serious.

6.4 Model Behavior in the Various Algorithms

The trends of the comparative performance of models are presented in Fig. 13, where there is little difference between the best-performing boosting models (LightGBM, CatBoost, XGBoost) and significant differences with Logistic Regression.

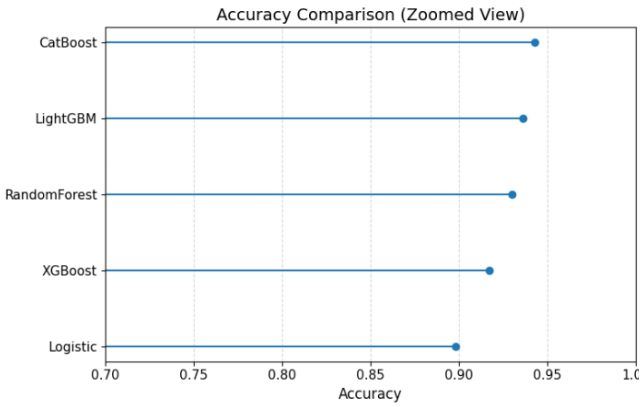

Fig.13. Accuracy Comparision

The accuracy comparison illustrated in Fig. 13 highlights the performance differences among the evaluated models. CatBoost achieves the highest accuracy, closely followed by LightGBM and Random Forest, indicating the strong capability of gradient boosting–based ensemble methods in capturing complex clinical patterns. XGBoost also demonstrates competitive performance, though slightly lower than the top-performing models. In contrast, Logistic Regression records the lowest accuracy, reflecting its limitation in modeling non-linear relationships inherent in heterogeneous clinical data. Overall, the results emphasize that advanced ensemble techniques consistently outperform traditional linear models, with marginal differences among the top models suggesting stable and robust predictive performance across the dataset.

6.5 Effect of Temporal Feature Removal

The effect of the elimination of the time characteristics is depicted in Fig. 14, in which the performance metrics of the model are evidently deteriorated. The significance of the temporal dynamics is indicated by the decrease in AUC.

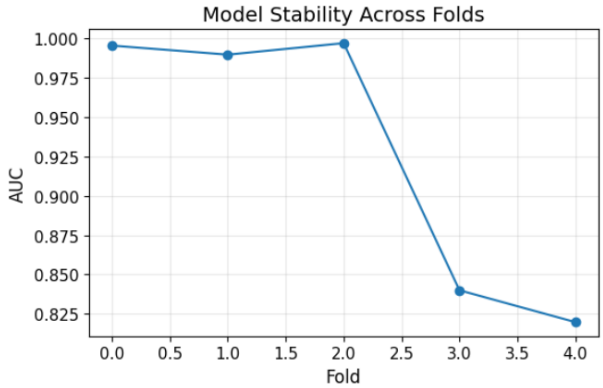

Fig.14. Model Stability Across Folds

This proves the fact that the disease progression is time-dependent in nature and loss of predictive signals vital to the disease progression is as a result of neglecting time.

7. CONCLUSION

This study proposes a novel Causal-Aware Conformal Prediction (CACP) framework for the early prediction of complicated pediatric appendicitis by integrating temporal feature engineering, causal modeling, ensemble learning, and uncertainty quantification. The approach addresses key limitations in traditional clinical prediction systems, particularly the lack of temporal progression awareness and reliable confidence estimation. Experimental results demonstrate strong predictive performance, achieving an AUC in the range of **0.97–0.98**, along with high accuracy and robustness across models. Among the evaluated methods, CatBoost and LightGBM show superior discriminative ability, while Random Forest achieves high recall, emphasizing the model's capability to effectively identify high-risk cases.

The inclusion of temporal velocity-based features significantly enhances predictive performance by capturing the dynamic progression of inflammatory markers, while causal modeling ensures that selected features maintain clinically meaningful relationships with disease severity, thereby improving interpretability. A major contribution of this work is the integration of conformal prediction, which enables the explicit identification of uncertain cases. This uncertainty-aware mechanism enhances clinical safety by flagging ambiguous instances for further evaluation rather than forcing potentially unreliable predictions. Additionally, calibration and error distribution analyses indicate that the model provides reliable probability estimates, with most misclassifications occurring near decision boundaries.

Overall, the proposed framework successfully balances accuracy, interpretability, and reliability, making it highly suitable for real-world clinical decision support. It not only generates predictions but also provides actionable insights such as risk trajectories, enabling timely intervention for high-risk patients where delays may lead to severe complications. Future work can focus on multi-center validation, integration with real-time clinical systems, and extension to other disease domains. Incorporating multimodal data sources such as imaging and longitudinal health records could further enhance predictive performance and broaden the applicability of the proposed approach.

Data Availability

The dataset used and/or analysed during the current study is publicly available on Kaggle at:

https://www.kaggle.com/datasets/joebeachcapital/regensburg-pediatric-appendicitis/data.

Additional processed data or analysis details are available from the corresponding author upon reasonable request.

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